

The Taylor tree in the paper's normalisation and the leading log coefficient

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Abstract

Two paper-facing corollaries of `twoD_taylor_tree_equal`. First, the substitution $N = \sqrt{n}$: for every T , $Z(\sqrt{n}) - \sum_{\alpha < 2T} n^{-\alpha/2} (A_\alpha \frac{\log n}{2} + B_\alpha) = O(n^{-T}(1 + \log n))$, so the paper's polynomial is $P_\mu(X) = \frac{1}{2} A_{2\mu} X + B_{2\mu}$. Second, the leading pole $\alpha = p$ is the collision of $i = j = 0$, the constant coefficient of the amplitude array is $c_{00}(s) = e^{\beta s x_{00}} y_{00}$, and therefore the coefficient of $N^{-p} \log N$ is $A_p = \frac{y_{00}}{k_1 k_2} \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s x_{00}} ds$: it depends on ξ only through $\xi(0)$ and on η only through $\eta(0)$. Zero `sorry/axiom`.

1 The things formalised

```
theorem twoD_taylor_tree_equal_n ... (T : ) :
  (fun n : => twoDamp b (Real.sqrt n) h h k k (anaAmp (ampCoeff x y) b)
    - polesBelow h h k k p T,
    n ^ (-( / 2)) * (canonA h h k k (fun i j s => ampCoeff x y (i, j) s)
      * (Real.log n / 2)
      + canonB b h h k k (anaFaceU (ampCoeff x y) b)
      (anaFaceV (ampCoeff x y) b) (fun i j s => ampCoeff x y (i, j) s) ))
  = 0[atTop] fun n : => n ^ (-T) * (1 + Real.log n)
theorem ampCoeff_zero ( : ) (x y : x → ) (s : ) :
  ampCoeff x y (0, 0) s = Real.exp ( * s * x (0, 0)) * y (0, 0)
theorem leading_pole_mem ... (hT : p < 2 * T) : p polesBelow h h k k p T
theorem le_of_mem_polesBelow ... ( : ) (h : polesBelow h h k k p T) : p
theorem canonC_leading ... (c : → → → ) :
  canonC h h k k c p = fun s => c 0 0 s / ((k : ) * k)
theorem leading_log_coeff ( : ) (h h k k : ) (hk : 0 < k) (hk : 0 < k) (p : )
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p) (x y : x → ) :
  canonA h h k k (fun i j s => ampCoeff x y (i, j) s) p
  = y (0, 0) / ((k : ) * k)
  * logMoment p 0 (fun s => Real.exp ( * s * x (0, 0)))
```

2 Mathematics

Composition with $n \mapsto \sqrt{n} \rightarrow \infty$ transports the N -statement; $(\sqrt{n})^{-\alpha} = n^{-\alpha/2}$ and $\log \sqrt{n} = \frac{1}{2} \log n$ identify the terms, and $(\sqrt{n})^{-2T}(1 + \frac{1}{2} \log n) \leq n^{-T}(1 + \log n)$ for $n \geq 1$ absorbs the remainder. For the leading coefficient, the box below $(0, 0)$ is the singleton, so $(y * E)_{00} = y_{00} E_{00}$, and $E_{00}(t) = \sum_{n \leq 0} t^n / n! (a^{*n})_{00} = \delta_{00} = 1$; the collision coefficient at $p = \alpha_0 = \delta_0$ is $c_{00}/(k_1 k_2)$ and its $M[p; 0; \cdot]$ moment is the displayed Gaussian integral. All poles are $\geq p$ since $i/k_1, j/k_2 \geq 0$.

3 Paper-facing meaning

The first result is `thm:TaylorTree` for $d = 2$ (equal starts) written literally in the paper’s variable n with $\mu = \alpha/2 \in \Lambda(h, k)$. The second is the $d = 2$ analogue of the leading-coefficient formula: the $n^{-p/2} \log n$ coefficient is $\frac{1}{2}A_p = \frac{\eta(0)}{2k_1k_2} \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s \xi(0)} ds$, a Gaussian moment of the fluctuation value at the origin. Astra #8 warned, correctly, that this “depends only on $\xi(0)$ ” property does NOT extend to B_p (which involves the face finite parts, hence ξ, η along the axes) nor to unequal starting exponents.

4 Lean notes

- `tendsto_rpow_atTop` with `Real.sqrt_eq_rpow` gives $\sqrt{\cdot} \rightarrow \infty$ in two lines; `IsBig0.comp_tendsto` then transports the statement and `congr'` with an `eventually_ge_atTop 1` filter fixes the terms.
- Rewrite `Real.log_sqrt` BEFORE `Real.sqrt_eq_rpow`: the latter rewrites every $\sqrt{n}$, including the one under the logarithm.
- `congr 1` on the subtraction keeps the $\sqrt{n}$ inside `twoDamp` untouched while the sum is rewritten term by term.
- `box (0, 0) = {(0, 0)}` by `ext` and `simp [mem_box, Prod.ext_iff, Nat.le_zero]`; `simp [convPow, delta]` evaluates E_{00} .
- Both files of this unit compiled on the first attempt.